

The Utility Problem

AMI (Advanced Metering Infrastructure) and smart meters are devices used for monitoring electricity usage in real time. These meters typically communicate data to the utility company using wireless communication systems, allowing for more accurate and efficient billing and grid management.

Smart meter flags are data points in the AMI/Smart Meter system that are used to indicate specific conditions or events that occur on the electricity grid which are “visible” to the particular AMI/smart meter. These flags can be used to identify and track issues such as power outages, voltage fluctuations, equipment failures and others.

Utilities use a variety of methods to detect energy theft, including the analysis of electricity usage patterns, the monitoring of voltage levels, and by utilizing of smart meter flags (these are, for example, reverse energy flags, reverse energy registers, tilt switches, blink counter - no power to the meter, etc.). When energy theft is detected, the utility company can use the energy theft flag to track the issue and take appropriate action, such as inspecting the customer's electricity supply or other legal action.

It is important to have an analytical system to analyze the smart meter flags because this will enable utilities to detect patterns and anomalies in the data that may indicate energy theft. Energy theft can result in significant financial losses for the utility company and can also create safety hazards. By detecting and preventing energy theft, utilities can improve the efficiency and reliability of the electricity grid, reduce energy waste, and ensure that all customers are paying for the electricity they use.

The Data Solutions and Methods

The Awesense Energy Transition Platform integrates all metering devices in the grid. It is no different for AMI/smart meters. For each meter, information about set alerts is stored. This setup is very simple and is created through our True Grid Intelligence (TGI) visualization environment. This can be, for example, a voltage loss alert, meter tampering information (as mentioned above), and many more. All notifications can be displayed directly in this interface, for example, on the Grid Alerts page:

- In the tabular form where all attributes of alerts are listed. Additional information includes data such as element ID and many others;
- On a map, where the size of the circles on the map indicates the number of alerts at a location.

All alerts for a sensor can also be viewed directly in TGI on the Sensor Alerts page

If these tools are not sufficient, it is possible to create a completely new tool for such analyses. The Awesense Energy Transition Platform Energy Data Model (EDM) APIs help a lot with this as it integrates a wide variety of data. All the necessary data is available in one place (the data is

AMI/Smart Meter Flag Analysis for Theft Detection

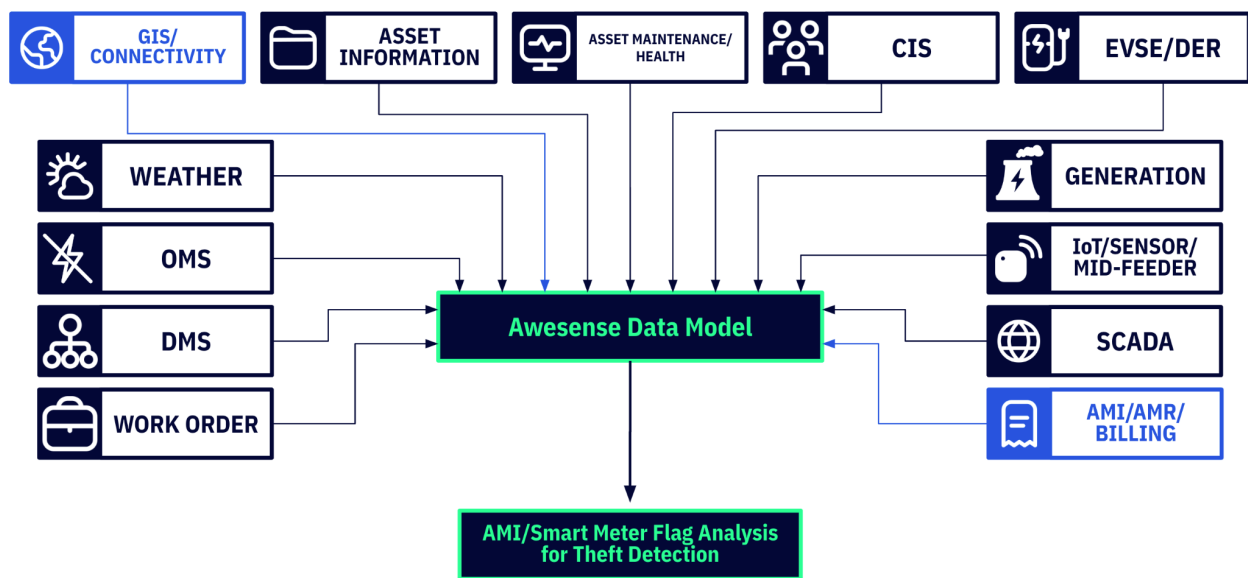
[UC48-01] - Awesense Energy Transition Platform Use Case



cleaned and synchronized), and is easy to access this data thanks to sophisticated functions for, for example, grid tracing and exports based on this tracing or time series data aggregation.

Required Data Sets

The required data for this use case are visualized in the following diagram. Data sources highlighted in [royal blue](#) are required for this use case.



Available Use Case Implementations:

Users of the Awesense Energy Transition Platform can access this use case by the following means:

1. TGI Grid Alerts and Sensor Alerts - Please speak to Awesense about access to the sandbox environment.
2. A GitHub link to the Jupyter notebook can be provided upon request.